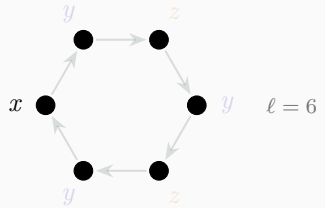
An Exponential Succinctness Gap between Three-Variable Logic and the Calculus of Relations

Yuya Uezato Yoshiki Nakamura
CyberAgent, Inc. Chiba University

STOC 2027 submission (author video for the PC)

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

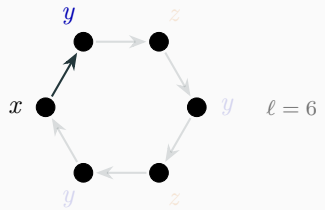
☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

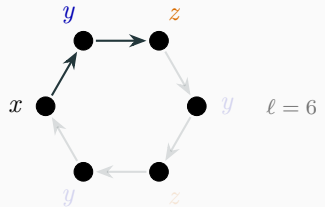
☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

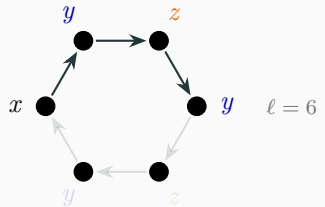
☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

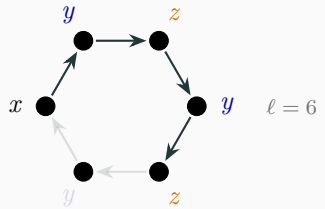
☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

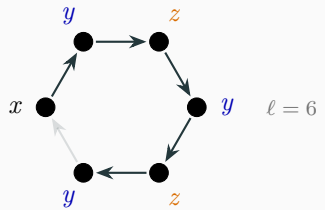
☞ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

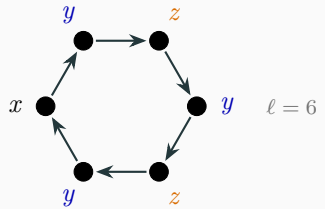
☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$\ell = 6$

$$\exists y\,(E(x,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge \exists z\,(E(y,z) \wedge \exists y\,(E(z,y) \wedge E(y,x))))))$$

Model checking: $\mathcal{M} \models \varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a, b \in \mathcal{M}$)

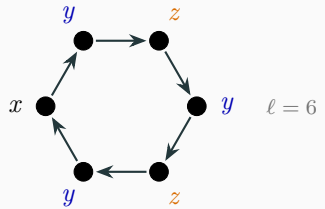
☛ $O(|\varphi| \cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi| n^k)$

k-variable logic FO^k

FO^k: first-order logic with k variables, **reusable**.
For example, "a closed walk of length ℓ at x" is expressible in FO^3:



$$\exists y\left(E(x,y)\wedge\exists z\left(E(y,z)\wedge\exists y\left(E(z,y)\wedge\exists z\left(E(y,z)\wedge\exists y\left(E(z,y)\wedge E(y,x)\right)\right)\right)\right)\right)$$

Model checking: $\mathcal{M}\models\varphi(a,b)$? (given $\mathcal{M}$ with n elements, a formula φ, and $a,b\in\mathcal{M}$)

👉 $O(|\varphi|\cdot n^k)$ for FO^k by table memoization [Var95, PODS]

ページ概要：

- FO^k は変数を使い回せる（長さ ℓ の walk も 3 変数）
- モデル検査は $O(|\varphi|n^k)$

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z.\ \cdots \wedge \cdots$, we use DNF translation.
$\rightsquigarrow$ $2^{O(|\varphi|)}$ compositions are used [Nak22].

ページ概要：

- $k \geq 4$ では n^k が最適、FO^3 だけ n^ω
- 代償：CoR への翻訳で合成が $2^{O(|\varphi|)}$ 個（∃ の内側の DNF 化）

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z. - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

ページ概要：

- $k \geq 4$ では n^k が最適、FO^3 だけ n^ω
- 代償：CoR への翻訳で合成が $2^{O(|\varphi|)}$ 個（∃ の内側の DNF 化）

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z. - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

ページ概要：

- $k \geq 4$ では n^k が最適、FO^3 だけ n^ω
- 代償：CoR への翻訳で合成が $2^{O(\lvert\varphi\rvert)}$ 個（∃ の内側の DNF 化）

FO^k model checking: $O(n^k)$ is 'optimal' for $k \geq 4$; the exception is FO^3

We consider binary relation structures with n elements:

syntax	model checking	
FO^k, $k \geq 4$	no $O(n^{k-\varepsilon})$ under HyperClique Hypothesis	[BS21, ICALP] [NU27, PODS]
FO^3	$2^{O(\lvert\varphi\rvert)} \cdot n^\omega$ no $2^{o(\lvert\varphi\rvert)} \cdot n^{\omega-\varepsilon}$ under 3-OV Hypothesis	[NU27, PODS] [Wil14, CSL-LICS, for PNF] [NU27, PODS]
CoR [Tar41, JSL] Tarski's calculus of relations	$O(\lvert\varphi\rvert \cdot n^\omega)$	

Boolean matrix algorithm of FO^3 model checking via $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ [Tar41, JSL; TG87]:

$$\exists z.\ P(x,z) \wedge Q(z,y) \quad \rightsquigarrow \quad (P;Q)(x,y)$$

composition = Boolean matrix multiplication

To obtain the form of $\exists z. - \wedge -$, we use DNF translation.
$\rightsquigarrow$ $2^{O(\lvert\varphi\rvert)}$ compositions are used [Nak22].

ページ概要：

- $k \geq 4$ では n^k が最適、FO^3 だけ n^ω
- 代償：CoR への翻訳で合成が $2^{O(\lvert\varphi\rvert)}$ 個（∃ の内側の DNF 化）

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y)$$

(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 *to* CoR*?*

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

ページ概要：

- この指数個の合成は必要か？
- 未解決（existential CoR のみ既知）

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \big(\bigcap_{i \in A} P_i \big) ; \big(\bigcap_{i \notin A} Q_i \big) \Big)(x,y)$$

(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

ページ概要：

- この指数個の合成は必要か？
- 未解決（existential CoR のみ既知）

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\begin{aligned} & \exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big) \\ \rightsquigarrow_{\text{(DNF)}} \ & \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big) \\ \rightsquigarrow_{\text{(to CoR)}} \ & \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y) \end{aligned}$$

(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

ページ概要：

- この指数個の合成は必要か？
- 未解決（existential CoR のみ既知）

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y)$$

(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

ページ概要：

- この指数個の合成は必要か？
- 未解決（existential CoR のみ既知）

Are $2^{O(|\varphi|)}$ compositions unavoidable?

DNF translation is used, for example:

$$\exists z.\ \big(P_1(x,z) \vee Q_1(z,y)\big) \wedge \cdots \wedge \big(P_k(x,z) \vee Q_k(z,y)\big)$$

$$\rightsquigarrow_{\text{(DNF)}}\ \exists z.\ \bigvee_{A \subseteq [k]} \Big(\bigwedge_{i \in A} P_i(x,z) \wedge \bigwedge_{i \notin A} Q_i(z,y) \Big)$$

$$\rightsquigarrow_{\text{(to CoR)}}\ \Big(\bigcup_{A \subseteq [k]} \Big(\bigcap_{i \in A} P_i \Big) ; \Big(\bigcap_{i \notin A} Q_i \Big) \Big)(x,y)$$

(In this translation, we use 2^k compositions, since the number of subsets $A \subseteq [k]$ is 2^k.)

Problem (Open [FGL+15, Inf. Sci.; Nak22, JLAMP; NU27, PODS])

Can we prove that there is no polynomial-size translation from FO^3 to CoR?

Partial answer [Nak22, JLAMP]: yes for **existential** CoR [Nak23, LICS] ($\equiv$ existential FO^3), via preservation under **extensions**, which fails for full CoR.

ページ概要：

- この指数個の合成は必要か？
- 未解決（existential CoR のみ既知）

Outline

1. Our results

2. Technical overview

3. Conclusion

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR.
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with TC

👉 Our proof technique can apply to all the above!

ページ概要：

- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
- 同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR*.*
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with TC

👉 Our proof technique can apply to all the above!

ページ概要：

- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
- 同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR*.*
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with TC

👉 Our proof technique can apply to all the above!

ページ概要：

- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
- 同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR*.*
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with TC

👉 Our proof technique can apply to all the above!

ページ概要：

- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
- 同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界

Our results

The answer is **YES!**

Thm. (This paper)

There is no polynomial-size translation from FO^3 *to* CoR*.*
More precisely, there are FO^3 *formulas* $\varphi_1, \varphi_2, \ldots$ *with* $|\varphi_k| = O(k)$ *such that every* CoR *term equivalent to* φ_k *needs* $\geq 2^{k-1}$ *compositions.*

Also in this paper: exponential lower bounds for

- $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$, binary — before:
  exponential gap FO^k vs. FO^{k-1} ($k \geq 4$), higher arity [VS24, LICS]
  $\mathrm{FO}^3 \rightsquigarrow \mathrm{FO}^2$ is polynomial over linear orders [GS05, LMCS]
- ARA(3) $\rightsquigarrow$ matrix circuits — asked by [BGV19, arXiv; BGV22, AMAI]
- $\mathrm{FO}^3 \rightsquigarrow \mathrm{CoR}$ with TC

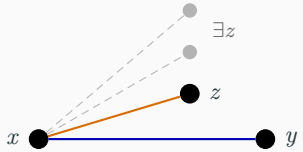
👉 Our proof technique can apply to all the above!

ページ概要：

- 答えは YES：$|\varphi_k| = O(k)$ で合成 $\geq 2^{k-1}$ 個
- 同じ手法で FO^2、行列言語、閉包付き CoR にも指数下界

The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



$G_k(x,y)$ denotes that there exists a z such that
for every $i \in [k]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order.

CoR: 2^k compositions

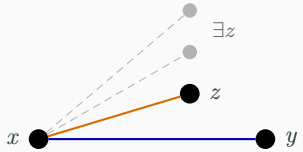
$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

ページ概要：

- G_k：各 i で x-y 辺と x-z 辺の一方が P_i、他方が Q_i
- DNF で合成 2^k 個、これは 2 倍を除いて最適

The hard formula G_k

$$G_k(x,y) \coloneqq \exists z \bigwedge_{i=1}^{k} \left[(P_i(x,y) \wedge Q_i(x,z)) \vee (Q_i(x,y) \wedge P_i(x,z)) \right] \qquad (|G_k| = O(k))$$



$G_k(x,y)$ denotes that there exists a z such that
for every $i \in [k]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order.

CoR: 2^k **compositions**

$$G_k \equiv \bigcup_{A \subseteq [k]} \left[\left(\bigcap_{i \in A} P_i \cap \bigcap_{i \notin A} Q_i \right) \cap \left(\left(\bigcap_{i \in A} Q_i \cap \bigcap_{i \notin A} P_i \right) ; \mathbf{1} \right) \right]$$

ページ概要：

- G_k：各 i で x-y 辺と x-z 辺の一方が P_i、他方が Q_i
- DNF で合成 2^k 個、これは 2 倍を除いて最適

Outline

1. Our results

2. Technical overview

3. Conclusion

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

☞ if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

☞ if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

☞ if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

👉 if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

👉 if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

☛ if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Proof outline (at high-level)

Goal Lemma Every CoR term equivalent to G_k needs $2^{\Omega(k)}$ compositions.

We construct structures $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ such that

- G_k distinguishes all of $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$:

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
G_k	T	F	F	$\cdots$	F	2^k	$\geq 2^k/2$

- **Key Lemma** If a CoR term t distinguishes d structures among $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ from $\mathcal{M}_0$, then t needs $\geq d/2$ compositions.

	$\mathcal{M}_0$	$\mathcal{M}_1$	$\mathcal{M}_2$	$\mathcal{M}_3$	$\mathcal{M}_4$	$\mathcal{M}_5$	$\cdots$	$\mathcal{M}_{2^k}$	#diff.	#comp.
t	T	F	T	F	T	T	$\cdots$	T	2	≥ 1
t'	F	T	T	T	T	F	$\cdots$	F	4	≥ 2

👉 if $t \equiv G_k$, then t needs $\geq 2^k/2$ compositions.

TASK How can we construct such $\mathcal{M}_0$ and $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$?

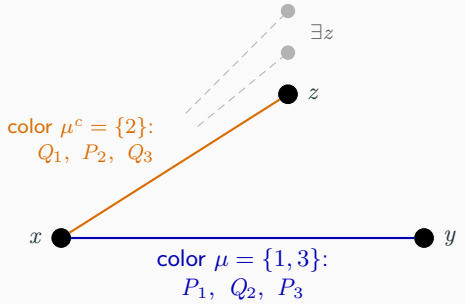
ページ概要：

- G_k が $\mathcal{M}_0$ で成り立ち、$\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$ で成り立たない構造を作る
- Key Lemma：d 個を $\mathcal{M}_0$ から見分ける項は合成 $\geq d/2$ 個

Structures as colored complete graphs

View structures as **complete graphs**, colored with 2^k colors $\mu \subseteq [k]$: $P_i \iff i \in \mu$.

Q_i is used as the complement of P_i (i.e., $i \notin \mu$): exactly one of P_i, Q_i holds on each edge.



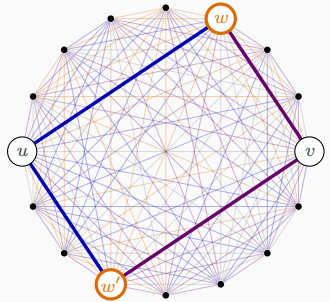
for every $i \in [3]$,
the x-y and x-z edges
have labels P_i and Q_i, in either order

☛ $G_k(x, y)$ holds iff some x-z edge has the color μ^c, where μ is the color of the x-y edge.

ページ概要：

- 構造を色付き完全グラフとして見る
- $G_k(x, y)$：x-y 辺の色の補の色の x-z 辺がある

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k4^k)$ vertices,
cf. [JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** w such that the u-w and w-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

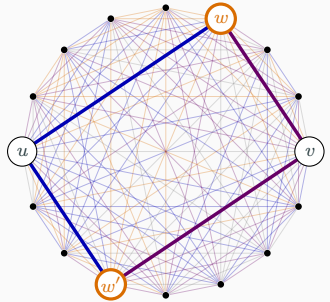
Every CoR term t is color-invariant on $\mathcal{M}_0$: $\mathcal{M}_0 \models t(u,v) \iff \mathcal{M}_0 \models t(u',v')$ whenever the u-v and u'-v' edges have the same color.

Proof by induction on t: the case, $t = t_1 \,;\, t_2$, follows from the (two-)witness property.

ページ概要：

- $\mathcal{M}_0$：ランダム彩色、two-witness property
- だからどの項も color-invariant（帰納法、合成の場合に two-witness）

The randomly colored structure $\mathcal{M}_0$



$\mathcal{M}_0$: randomly colored, $O(k\,4^k)$ vertices,
cf. [JMT95, Algebra Univers.]

Two-witness property For all $u \neq v$ and all colors $\boldsymbol{\mu}, \boldsymbol{\nu}$, there are at least two **witnesses** w such that the u-w and w-v edges have colors $\boldsymbol{\mu}$ and $\boldsymbol{\nu}$, respectively.

Every CoR term t is color-invariant on $\mathcal{M}_0$: $\mathcal{M}_0 \models t(u, v) \iff \mathcal{M}_0 \models t(u', v')$ whenever the u-v and u'-v' edges have the same color.

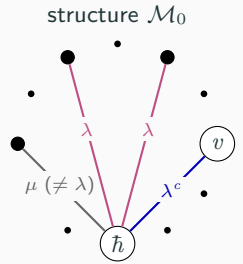
Proof by induction on t: the case, $t = t_1 \,;\, t_2$, follows from the (two-)witness property.

ページ概要：

- $\mathcal{M}_0$：ランダム彩色、two-witness property
- だからどの項も color-invariant（帰納法、合成の場合に two-witness）

$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Fix an arbitrary vertex $\hbar$ of $\mathcal{M}_0$, the **hub**.



$\hbar$ has λ-edges

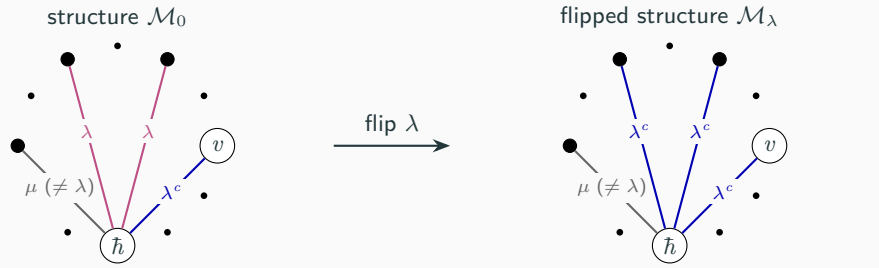
For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).

ページ概要：

- hub の λ-edges を λ^c に flip $\to$ hub に λ-edges がない
- $\mathcal{M}_0 \models G_k(\hbar, v)$ だが $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$
- 色ごとに 1 つ、計 2^k 個

$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Fix an arbitrary vertex $\hbar$ of $\mathcal{M}_0$, the **hub**.



$\hbar$ has λ-edges

$\hbar$ does not have λ-edges

(Edges not at $\hbar$ are unchanged from $\mathcal{M}_0$.)

For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.

Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.

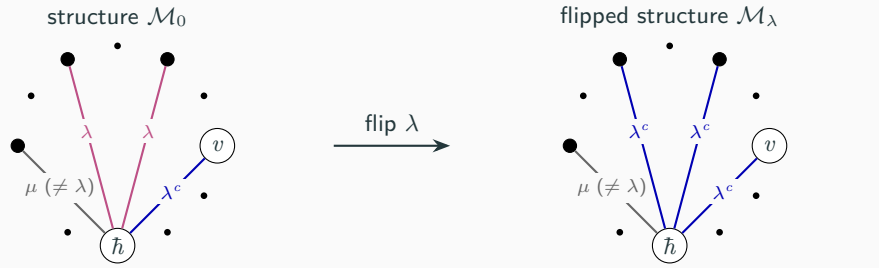
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).

ページ概要：

- hub の λ-edges を λ^c に flip $\rightarrow$ hub に λ-edges がない
- $\mathcal{M}_0 \models G_k(\hbar, v)$ だが $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$
- 色ごとに 1 つ、計 2^k 個

$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Fix an arbitrary vertex $\hbar$ of $\mathcal{M}_0$, the **hub**.



$\hbar$ has λ-edges

$\hbar$ does not have λ-edges

(Edges not at $\hbar$ are unchanged from $\mathcal{M}_0$.)

For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.

Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.

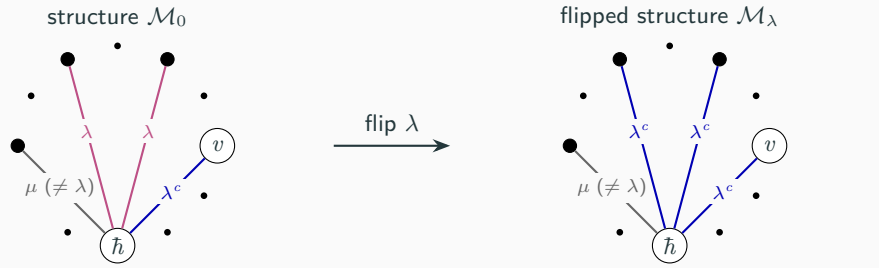
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).

ページ概要：

- hub の λ-edges を λ^c に flip $\rightarrow$ hub に λ-edges がない
- $\mathcal{M}_0 \models G_k(\hbar, v)$ だが $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$
- 色ごとに 1 つ、計 2^k 個

$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Fix an arbitrary vertex $\hbar$ of $\mathcal{M}_0$, the **hub**.



$\hbar$ has λ-edges

$\hbar$ does not have λ-edges

(Edges not at $\hbar$ are unchanged from $\mathcal{M}_0$.)

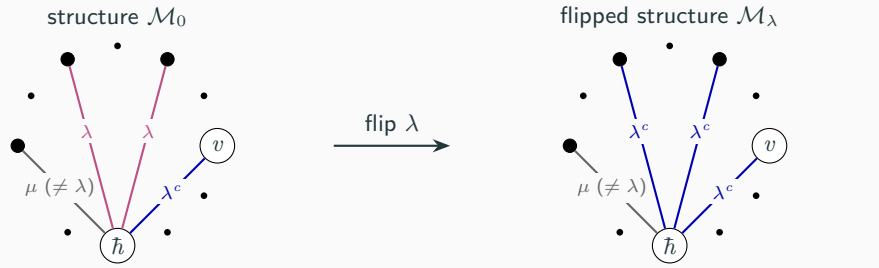
For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).

ページ概要：

- hub の λ-edges を λ^c に flip $\rightarrow$ hub に λ-edges がない
- $\mathcal{M}_0 \models G_k(\hbar, v)$ だが $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$
- 色ごとに 1 つ、計 2^k 個

$\mathcal{M}_\lambda$: flipping at the hub $\hbar$

Fix an arbitrary vertex $\hbar$ of $\mathcal{M}_0$, the **hub**.



$\hbar$ has λ-edges

$\hbar$ does not have λ-edges

(Edges not at $\hbar$ are unchanged from $\mathcal{M}_0$.)

For v with the $\hbar$-v edge of color λ^c: $\mathcal{M}_0 \models G_k(\hbar, v)$ but $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$.
Flipping each of the 2^k colors λ gives $\mathcal{M}_1, \ldots, \mathcal{M}_{2^k}$.
$\rightsquigarrow$ Each composition newly distinguishes at most two of them (next slides).

ページ概要：

- hub の λ-edges を λ^c に flip $\rightarrow$ hub に λ-edges がない
- $\mathcal{M}_0 \models G_k(\hbar, v)$ だが $\mathcal{M}_\lambda \not\models G_k(\hbar, v)$
- 色ごとに 1 つ、計 2^k 個

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

Consider $t\,;s$, where t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$.
Q. For which pairs (u, v) can $\mathcal{M}_0 \models (t\,;s)(u, v) \iff \mathcal{M}_\lambda \models (t\,;s)(u, v)$ fail?
t are s are color-invariant on both $\mathcal{M}_0$ and $\mathcal{M}_\lambda$:
w is a witness depends only on the colors (μ, ν) of the u-w and w-v edges.

Case $\hbar \notin \{u, v\}$: Never. (The equivalence is always preserved.)

ページ概要：

- t, s がまだ見分けていないとき、$t\,;s$ はどの点対で iff が崩れうるか
- $\mathcal{M}_0$ と $\mathcal{M}_\lambda$ の違いは hub の辺だけ
- $\hbar \notin \{u, v\}$：どの色の組 (μ, ν) でも hub 以外で witness を取り直せる（two witnesses）→ iff は保たれる

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

Consider $t\,;s$, where t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$.
Q. For which pairs (u, v) can $\mathcal{M}_0 \models (t\,;s)(u, v) \iff \mathcal{M}_\lambda \models (t\,;s)(u, v)$ fail?
t are s are color-invariant on both $\mathcal{M}_0$ and $\mathcal{M}_\lambda$:
w is a witness depends only on the colors (μ, ν) of the u-w and w-v edges.

Case $\hbar \notin \{u, v\}$: Never. (The equivalence is always preserved.)



ページ概要：

- t, s がまだ見分けていないとき、$t\,;s$ はどの点対で iff が崩れうるか
- $\mathcal{M}_0$ と $\mathcal{M}_\lambda$ の違いは hub の辺だけ
- $\hbar \notin \{u, v\}$：どの色の組 (μ, ν) でも hub 以外で witness を取り直せる（two witnesses）→ iff は保たれる

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

Consider $t\,;s$, where t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$.
Q. For which pairs (u, v) can $\mathcal{M}_0 \models (t\,;s)(u, v) \iff \mathcal{M}_\lambda \models (t\,;s)(u, v)$ fail?
t are s are color-invariant on both $\mathcal{M}_0$ and $\mathcal{M}_\lambda$:
w is a witness depends only on the colors (μ, ν) of the u-w and w-v edges.

Case $\hbar \notin \{u, v\}$: Never. (The equivalence is always preserved.)



The **two-witness property** shows
there is w with the same colors (μ, ν)

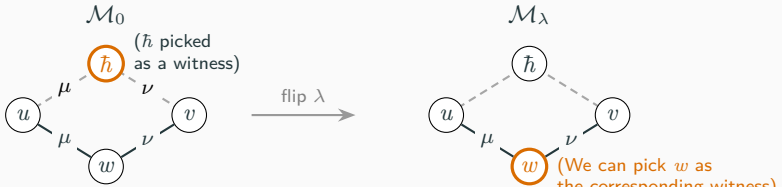
ページ概要：

- t, s がまだ見分けていないとき、$t\,;s$ はどの点対で iff が崩れうるか
- $\mathcal{M}_0$ と $\mathcal{M}_\lambda$ の違いは hub の辺だけ
- $\hbar \notin \{u, v\}$：どの色の組 (μ, ν) でも hub 以外で witness を取り直せる（two witnesses）→ iff は保たれる

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (1/2)

Consider $t\,;s$, where t and s do not distinguish $\mathcal{M}_\lambda$ from $\mathcal{M}_0$.
Q. For which pairs (u, v) can $\mathcal{M}_0 \models (t\,;s)(u, v) \iff \mathcal{M}_\lambda \models (t\,;s)(u, v)$ fail?
t are s are color-invariant on both $\mathcal{M}_0$ and $\mathcal{M}_\lambda$:
w is a witness depends only on the colors (μ, ν) of the u-w and w-v edges.

Case $\hbar \notin \{u, v\}$: Never. (The equivalence is always preserved.)



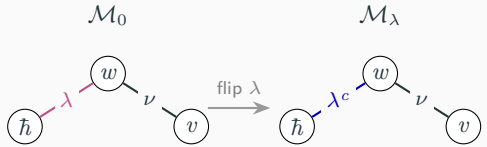
The **two-witness property** shows there is w with the same colors (μ, ν)

ページ概要：

- t, s がまだ見分けていないとき、$t\,;s$ はどの点対で iff が崩れうるか
- $\mathcal{M}_0$ と $\mathcal{M}_\lambda$ の違いは hub の辺だけ
- $\hbar \notin \{u, v\}$：どの色の組 (μ, ν) でも hub 以外で witness を取り直せる（two witnesses）→ iff は保たれる

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (2/2)

Case $\hbar \in \{u, v\}$, say $u = \hbar$: the edge $\hbar$-w itself may be flipped. ($v = \hbar$: the same with s; $u = v = \hbar$: similar)



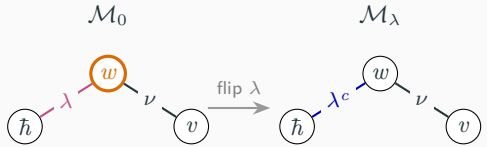
$\mathcal{M}_\lambda$ is newly distinguished only if t or s holds only on λ-edges
$\Rightarrow$ at most two such λ (one for t, one for s).

ページ概要：

- $\hbar \in \{u, v\}$（例えば $u = \hbar$）：t が λ-edges だけで成り立つときだけ変わる（s も同様）
- だから各合成が新たに見分ける $\mathcal{M}_\lambda$ は高々 2 個
- $(\hbar, \hbar)$ の場合も高々 2 は変わらない（論文参照）

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (2/2)

Case $\hbar \in \{u, v\}$, say $u = \hbar$: the edge $\hbar$-w itself may be flipped. ($v = \hbar$: the same with s; $u = v = \hbar$: similar)



if t holds **only** on λ-edges:
no witness in $\mathcal{M}_\lambda$

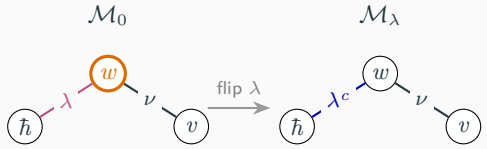
$\mathcal{M}_\lambda$ is newly distinguished only if t or s holds only on λ-edges
$\Rightarrow$ at most two such λ (one for t, one for s).

ページ概要：

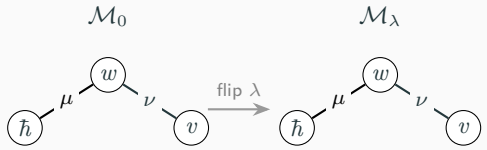
- $\hbar \in \{u, v\}$（例えば $u = \hbar$）：t が λ-edges だけで成り立つときだけ変わる（s も同様）
- だから各合成が新たに見分ける $\mathcal{M}_\lambda$ は高々 2 個
- $(\hbar, \hbar)$ の場合も高々 2 は変わらない（論文参照）

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (2/2)

Case $\hbar \in \{u, v\}$, say $u = \hbar$: the edge $\hbar$-w itself may be flipped. ($v = \hbar$: the same with s; $u = v = \hbar$: similar)



if t holds **only** on λ-edges:
no witness in $\mathcal{M}_\lambda$



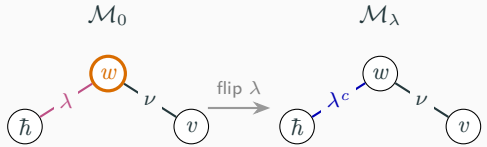
$\mathcal{M}_\lambda$ is newly distinguished only if t or s holds only on λ-edges
$\Rightarrow$ at most two such λ (one for t, one for s).

ページ概要：

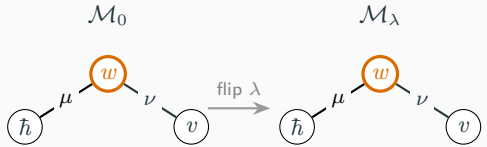
- $\hbar \in \{u, v\}$（例えば $u = \hbar$）：t が λ-edges だけで成り立つときだけ変わる（s も同様）
- だから各合成が新たに見分ける $\mathcal{M}_\lambda$ は高々 2 個
- $(\hbar, \hbar)$ の場合も高々 2 は変わらない（論文参照）

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (2/2)

Case $\hbar \in \{u, v\}$, say $u = \hbar$: the edge $\hbar$-w itself may be flipped. ($v = \hbar$: the same with s; $u = v = \hbar$: similar)



if t holds **only** on λ-edges:
no witness in $\mathcal{M}_\lambda$



if t also holds on a color $\mu \neq \lambda$:
a μ-edge at $\hbar$ is not flipped

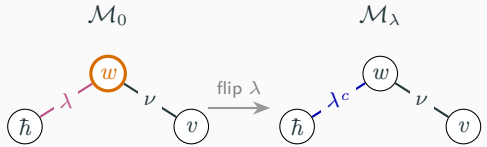
$\mathcal{M}_\lambda$ is newly distinguished only if t or s holds only on λ-edges
$\Rightarrow$ at most two such λ (one for t, one for s).

ページ概要：

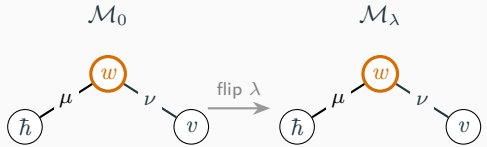
- $\hbar \in \{u, v\}$（例えば $u = \hbar$）：t が λ-edges だけで成り立つときだけ変わる（s も同様）
- だから各合成が新たに見分ける $\mathcal{M}_\lambda$ は高々 2 個
- $(\hbar, \hbar)$ の場合も高々 2 は変わらない（論文参照）

Why a composition newly distinguishes at most two $\mathcal{M}_\lambda$ (2/2)

Case $\hbar \in \{u, v\}$, say $u = \hbar$: the edge $\hbar$-w itself may be flipped. ($v = \hbar$: the same with s; $u = v = \hbar$: similar)



if t holds **only** on λ-edges:
no witness in $\mathcal{M}_\lambda$



if t also holds on a color $\mu \neq \lambda$:
a μ-edge at $\hbar$ is not flipped

$\mathcal{M}_\lambda$ is newly distinguished only if t or s holds only on λ-edges
$\Rightarrow$ at most two such λ (one for t, one for s).

ページ概要：

- $\hbar \in \{u, v\}$（例えば $u = \hbar$）：t が λ-edges だけで成り立つときだけ変わる（s も同様）
- だから各合成が新たに見分ける $\mathcal{M}_\lambda$ は高々 2 個
- $(\hbar, \hbar)$ の場合も高々 2 は変わらない（論文参照）

Outline

1. Our results

2. Technical overview

3. Conclusion

Conclusion

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $2^{\Omega(k)}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k flips of one structure $\mathcal{M}_0$; each composition newly distinguishes ≤ 2 of them.

Problem (Open Questions)

- *a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?*
- *the exact number of compositions for G_k: between 2^{k-1} and 2^k?*

Thank you for your attention!

ページ概要：

- 結果：合成 $2^{\Omega(k)}$ 個が必要；手法：$\mathcal{M}_0$ の 2^k 通りの flip
- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

Conclusion

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $2^{\Omega(k)}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k flips of one structure $\mathcal{M}_0$; each composition newly distinguishes ≤ 2 of them.

Problem (Open Questions)

- *a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?*
- *the exact number of compositions for G_k: between 2^{k-1} and 2^k?*

Thank you for your attention!

ページ概要：

- 結果：合成 $2^{\Omega(k)}$ 個が必要；手法：$\mathcal{M}_0$ の 2^k 通りの flip
- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

Conclusion

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $2^{\Omega(k)}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k flips of one structure $\mathcal{M}_0$; each composition newly distinguishes ≤ 2 of them.

Problem (Open Questions)

- *a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?*
- *the exact number of compositions for G_k: between 2^{k-1} and 2^k?*

Thank you for your attention!

ページ概要：

- 結果：合成 $2^{\Omega(k)}$ 個が必要；手法：$\mathcal{M}_0$ の 2^k 通りの flip
- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

Conclusion

Result Tarski's translation from FO^3 to CoR is optimal in the worst case:

- some FO^3 formulas of size $O(k)$ need $2^{\Omega(k)}$ compositions in every equivalent CoR term, even with complement, converse, and shared subterms;
- the same technique: FO^3 vs. FO^2, matrix query languages, CoR with closure.

Technique 2^k flips of one structure $\mathcal{M}_0$; each composition newly distinguishes ≤ 2 of them.

Problem (Open Questions)

- *a superpolynomial gap over a **fixed** finite signature (e.g., one binary relation)?*
- *the domain size from which the blowup begins?*
- *the exact number of compositions for G_k: between 2^{k-1} and 2^k?*

Thank you for your attention!

ページ概要：

- 結果：合成 $2^{\Omega(k)}$ 個が必要；手法：$\mathcal{M}_0$ の 2^k 通りの flip
- 未解決：固定シグネチャ、開始の領域サイズ、正確な合成数

References i

[Var95] M. Y. Vardi. "On the Complexity of Bounded-Variable Queries". In: **Proceedings of the Fourteenth ACM SIGACT-SIGMOD-SIGART Symposium on Principles of Database Systems (PODS 1995)**. ACM, 1995, pp. 266–276.

[BS21] K. Bringmann and J. Slusallek. "Current Algorithms for Detecting Subgraphs of Bounded Treewidth Are Probably Optimal". In: **48th International Colloquium on Automata, Languages, and Programming (ICALP)**. Vol. 198. Leibniz International Proceedings in Informatics (LIPIcs). Schloss Dagstuhl – Leibniz-Zentrum für Informatik, 2021, 40:1–40:16. DOI: 10.4230/LIPIcs.ICALP.2021.40.

[NU27] Y. Nakamura and Y. Uezato. "A Fine-Grained Dichotomy for Bounded-Variable Query Evaluation: The Calculus of Relations, a Boolean Modal Logic, and One-Variable Counting Logic". In: **Proceedings of the ACM on Management of Data** 4.5 (PODS) (2027), 295:1–295:27. DOI: 10.1145/3850130.

[Wil14] R. Williams. "Faster Decision of First-Order Graph Properties". In: **Joint Meeting of the Twenty-Third EACSL Annual Conference on Computer Science Logic and the Twenty-Ninth Annual ACM/IEEE Symposium on Logic in Computer Science (CSL-LICS)**. ACM, 2014, 80:1–80:6. DOI: 10.1145/2603088.2603121.

References ii

[Tar41] A. Tarski. "On the Calculus of Relations". In: **The Journal of Symbolic Logic** 6.3 (1941), pp. 73–89. DOI: 10.2307/2268577.

[TG87] A. Tarski and S. Givant. **A Formalization of Set Theory without Variables**. Vol. 41. Colloquium Publications. Providence, RI: American Mathematical Society, 1987. DOI: 10.1090/coll/041.

[Nak22] Y. Nakamura. "Expressive Power and Succinctness of the Positive Calculus of Binary Relations". In: **Journal of Logical and Algebraic Methods in Programming** 127 (2022), p. 100760. DOI: 10.1016/j.jlamp.2022.100760.

[FGL+15] G. H. L. Fletcher, M. Gyssens, D. Leinders, D. Surinx, J. Van den Bussche, D. Van Gucht, S. Vansummeren, and Y. Wu. "Relative Expressive Power of Navigational Querying on Graphs". In: **Information Sciences** 298 (2015), pp. 390–406. DOI: 10.1016/j.ins.2014.11.031.

[Nak23] Y. Nakamura. "Existential Calculi of Relations with Transitive Closure: Complexity and Edge Saturations". In: **38th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)**. IEEE, 2023, pp. 1–13. DOI: 10.1109/LICS56636.2023.10175811.

References iii

[VS24] H. Vinall-Smeeth. "From Quantifier Depth to Quantifier Number: Separating Structures with k Variables". In: **Proceedings of the 39th Annual ACM/IEEE Symposium on Logic in Computer Science (LICS)**. Full version: arXiv:2311.15885. ACM, 2024, 71:1–71:14.

[GS05] M. Grohe and N. Schweikardt. "The Succinctness of First-Order Logic on Linear Orders". In: **Logical Methods in Computer Science** 1.1:6 (2005). Conference version in LICS 2004, pp. 1–25. DOI: 10.2168/LMCS-1(1:6)2005.

[BGV19] R. Brijder, M. Gyssens, and J. Van den Bussche. **On Matrices and K-Relations**. arXiv:1904.03934v1. 2019. arXiv: 1904.03934 [cs.DB]. URL: https://arxiv.org/abs/1904.03934v1.

[BGV22] R. Brijder, M. Gyssens, and J. Van den Bussche. "On Matrices and K-Relations". In: **Annals of Mathematics and Artificial Intelligence** 90.2–3 (2022), pp. 181–210. DOI: 10.1007/s10472-021-09760-4.

[JMT95] P. Jipsen, R. D. Maddux, and Z. Tuza. "Small Representations of the Relation Algebra $E_{n+1}(1, 2, 3)$". In: **Algebra Universalis** 33 (1995), pp. 136–139. DOI: 10.1007/BF01190770.